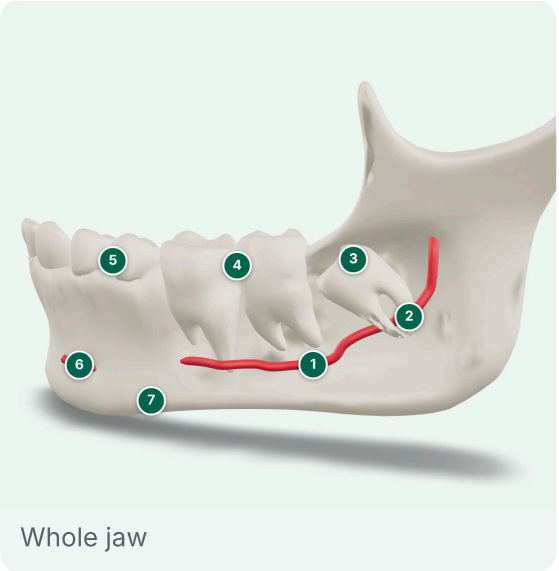
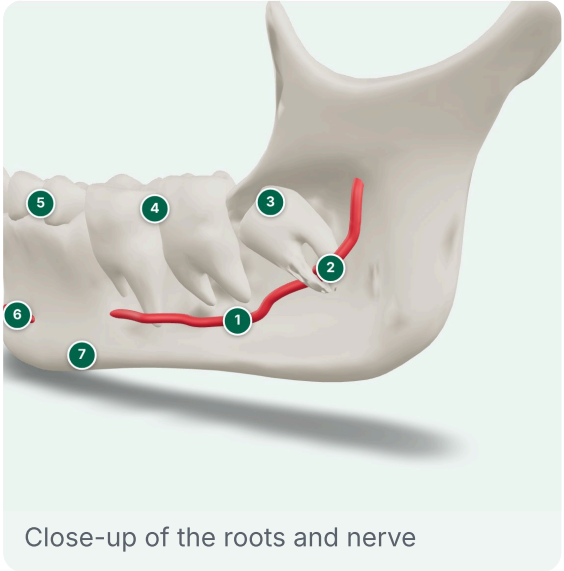
NAGOYA UNIVERSITY

Impacted lower wisdom tooth and the nerve near it



Whole jaw



Close-up of the roots and nerve

1. Jaw nerve (inferior alveolar nerve) 2. Roots of the wisdom tooth 3. Wisdom tooth (third molar)
4. First and second molars 5. First and second premolars 6. Chin nerve (mental nerve)
7. Jaw bone (mandible)

Jaw bone and teeth Nerve, artery and vein (in their canal)

What is the problem?

The wisdom tooth is the last tooth at the back. Its other name is the **third molar**. An **impacted lower wisdom tooth** is stuck under the gum and bone. It has not come through the gum (it is **unerupted**). It cannot grow in the right way.

The nerve

A nerve runs in a tube of bone inside the lower jaw. Here it is called the **jaw nerve**. Its medical name is the **inferior alveolar nerve**. Two blood vessels run with it in the same tube. They are the **inferior alveolar artery and vein**.[1]

The jaw nerve gives feeling to the lower teeth and gums on the same side.[2] Near the front of the jaw, a branch of it comes out of the bone through a small hole. This branch is the **chin nerve** (**mental nerve**). It gives feeling to the lower lip and chin.[3]

The tongue gets its feeling from a different nerve, the **lingual nerve**. It can also be hurt when a wisdom tooth is taken out.[4]

In this model, **the roots of the wisdom tooth wrap around the nerve and artery**.

Why taking the tooth out can hurt the nerve

To take the tooth out, the roots must come out of the bone. The roots curve around the nerve. So the nerve can be pressed, stretched or cut as the roots come out. The artery and vein can also be hurt and bleed.[1]

If the jaw nerve is hurt, the chin nerve may also stop working well. This is because it is a branch of the jaw nerve.[3]

What nerve damage feels like

The lower lip, chin, teeth or gums on that side may:[2]

- feel numb
- tingle, like pins and needles
- feel different, or strange
- hurt. Even a light touch may hurt.[4]

The lip and face can still move as normal.[2]

Short-term: the feeling comes back. This most often takes weeks to months.[2]
Long-term: the feeling does not fully come back.[2]

Age: the risk of nerve damage is higher in people over about 24 years old.[2] In older people, the feeling is also less likely to come back fully.[5] One small study followed people for 5 years. Everyone under 30 got full feeling back. Only 3 of 8 people over 30 did.[6] Studies do not agree on one exact age.

How often does it happen?

Some wisdom teeth have roots close to the nerve. A review of 16 studies looked at these teeth. It found:[7]

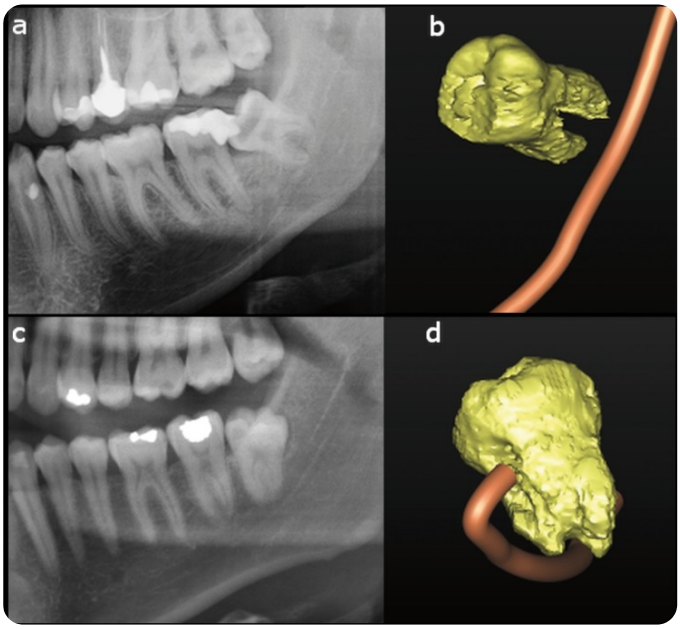
About 7 in 100
had nerve damage after the whole tooth was taken out

About 1 in 100
had nerve damage that did not go away

Why a 3D scan can help

The usual dental X-ray of the whole mouth is called an **OPG**. OPG is short for **orthopantomogram**. It is also called a panoramic X-ray. It is a flat picture. On a flat picture, the roots can look like they touch the nerve, even when they do not.[8] Some signs on an OPG can warn that the roots are close to the nerve.[9]

A **CBCT** scan is a 3D X-ray. CBCT is short for **cone beam computed tomography**. It can show if the roots really touch the nerve. It also shows which side of the roots the nerve is on.[8] Studies show it does this better than an OPG.[10]



The two OPGs (a and c) look much the same. The CBCT images show they are not. Top (b): the tooth does not touch the nerve. Bottom (d): the roots wrap around the nerve. In b and d, the tooth is yellow and the nerve is orange.[8]

Picture: Rodriguez y Baena R, et al. J Clin Exp Dent. 2017, Figure 1. Licence: CC BY 2.5. Not changed.

A CBCT scan uses more X-ray than an OPG. So it is used when an OPG shows the roots may be close to the nerve.[8]

Choices your dentist/surgeon might offer

- **Leave the tooth in and check it often**, if it is not causing problems.
- **Coronectomy:** take out only the top of the tooth. Leave the roots around the nerve. In the same review, about 1 in 200 people had nerve damage after this.[7] But the roots can move. Once in a while they need to come out later.
- **Take out the whole tooth.**

Questions you could ask your dentist/surgeon?

- How close are the roots to the nerve?
- What is the risk of nerve damage in this case?
- Is a CBCT scan needed?
- Is a coronectomy a good choice in this case?
- What happens if the tooth is left in?

Sources

1. Juodzbalys G, Wang HL, Sabalys G. Anatomy of mandibular vital structures. Part I: mandibular canal and inferior alveolar neurovascular bundle in relation with dental implantology. J Oral Maxillofac Res. 2010;1(1):e2. doi.org/10.5037/jomr.2010.1102
2. Sarikov R, Juodzbalys G. Inferior alveolar nerve injury after mandibular third molar extraction: a literature review. J Oral Maxillofac Res. 2014;5(4):e1. doi.org/10.5037/jomr.2014.5401
3. Juodzbalys G, Wang HL, Sabalys G. Anatomy of mandibular vital structures. Part II: mandibular incisive canal, mental foramen and associated neurovascular bundles in relation with dental implantology. J Oral Maxillofac Res. 2010;1(1):e3. doi.org/10.5037/jomr.2010.1103
4. Renton T, Yilmaz Z. Profiling of patients presenting with posttraumatic neuropathy of the trigeminal nerve. J Orofac Pain. 2011;25(4):333-344. PMID 22247929. pubmed.ncbi.nlm.nih.gov/22247929
5. Queral-Godoy E, Valmaseda-Castellón E, Berini-Aytés L, Gay-Escoda C. Incidence and evolution of inferior alveolar nerve lesions following lower third molar extraction. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 2005;99(3):259-264. doi.org/10.1016/j.tripleo.2004.06.001
6. Kjølle GK, Bjørnland T. Low risk of neurosensory dysfunction after mandibular third molar surgery in patients less than 30 years of age. A prospective study following removal of 1220 mandibular third molars. Oral Surg Oral Med Oral Pathol Oral Radiol. 2013;116(4):411-417. doi.org/10.1016/j.oooo.2013.06.024
7. Rico-Barroso L, Barroso-Medel M, Gutiérrez-Pérez J, Serrera-Figallo M, Torres-Lagares D. Coronectomy as an alternative technique to complete extraction of mandibular third molars with risk of nerve injury. Med Oral Patol Oral Cir Bucal. 2026;31(1):e19-e30. doi.org/10.4317/medoral.27459
8. Rodriguez y Baena R, Beltrami R, Tagliabo A, Rizzo S, Lupi SM. Differences between panoramic and Cone Beam-CT in the surgical evaluation of lower third molars. J Clin Exp Dent. 2017;9(2):e259-e265. doi.org/10.4317/jced.53234
9. Rood JP, Shehab BA. The radiological prediction of inferior alveolar nerve injury during third molar surgery. Br J Oral Maxillofac Surg. 1990;28(1):20-25. doi.org/10.1016/0266-4356(90)90005-6
10. Reia VCB, de Toledo Telles-Araujo G, Peralta-Mamani M, Biancardi MR, Rubira CMF, Rubira-Bullen IRF. Diagnostic accuracy of CBCT compared to panoramic radiography in predicting IAN exposure: a systematic review and meta-analysis. Clin Oral Investig. 2021;25(8):4721-4733. doi.org/10.1007/s00784-021-03942-4

This leaflet is for a research study at Nagoya University. It also gives general facts. It is not advice for one person. A dentist or oral surgeon can give advice for each person.